WORKED SOLUTION

STEP I 2017, Question 9

Mechanics • Projectile Motion

STEP I 2017, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

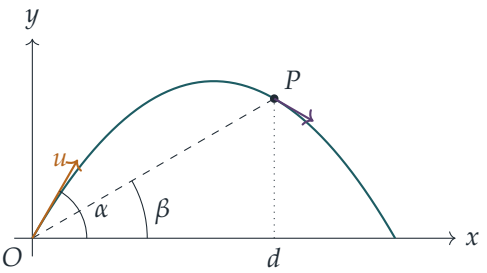
step.maths.org/questions/17-s1-q9

The key idea

Write down the trajectory equation and make it pass through P. That gives u^2 as a function of the projection angle alone, and the cleanest variable to work with is $T = \tan\alpha$, because $\sec^2\alpha = 1 + T^2$ turns everything into a rational function of T. Minimising that function is ordinary calculus, and the stationary-point equation turns out to be a double-angle formula in disguise.

Solution

Model the particle as moving freely under gravity alone (no air resistance), with O as origin, x horizontal towards P and y vertically up. Write $T = \tan\alpha$ and $k = \tan\beta > 0$.



Part (i)

With horizontal velocity $u\cos\alpha$, the particle reaches horizontal distance x at time $t = x/(u\cos\alpha)$, and its height then is

$$y = x\tan\alpha - \frac{gx^2}{2u^2\cos^2\alpha} = xT - \frac{gx^2(1+T^2)}{2u^2}.$$

Passing through $P = (d, dk)$ means

$$dk = dT - \frac{gd^2(1+T^2)}{2u^2} \quad \Longrightarrow \quad u^2 = \frac{gd}{2} \cdot \frac{1+T^2}{T-k}.$$

The right-hand side must be positive, so only angles with $T > k$, that is $\alpha > \beta$, are possible. That makes sense: we must aim above P. So we must minimise

$$f(T) = \frac{1+T^2}{T-k} \qquad (T > k).$$

By the quotient rule,

$$f'(T) = \frac{2T(T-k) - (1+T^2)}{(T-k)^2} = \frac{T^2 - 2kT - 1}{(T-k)^2}.$$

The numerator is zero at $T = k \pm \sqrt{k^2+1}$. Only the $+$ root exceeds k. Since $f(T) \to \infty$ both as $T \to k^+$ and as $T \to \infty$, and there is only one stationary point in between, that point gives the minimum. So the optimal α satisfies

$$T^2 - 2kT - 1 = 0. \qquad (*)$$

Ask yourself How do we simplify $f(T)$ at the minimum without substituting the surd?

Use $(*)$ to rewrite things. It gives $k = \dfrac{T^2-1}{2T}$, so

$$T - k = \frac{2T^2 - T^2 + 1}{2T} = \frac{1+T^2}{2T}, \qquad \text{hence} \qquad f(T) = \frac{1+T^2}{(1+T^2)/(2T)} = 2T.$$

Therefore

$$u^2 = \frac{gd}{2} \cdot 2T = gd\tan\alpha,$$

as required.

For the angle, $(*)$ in the form $k = \dfrac{T^2-1}{2T}$ reads

$$\tan\beta = -\frac{1-\tan^2\alpha}{2\tan\alpha} = -\cot 2\alpha = \tan(2\alpha - 90^\circ),$$

using $\tan 2\alpha = \dfrac{2\tan\alpha}{1-\tan^2\alpha}$ and $-\cot\theta = \tan(\theta - 90^\circ)$. (If $\alpha = 45^\circ$ then $(*)$ gives $k = 0$, which is excluded because $\beta > 0$, so $\cot 2\alpha$ is defined.) Since $0^\circ < \alpha < 90^\circ$, the angle $2\alpha - 90^\circ$ lies in $(-90^\circ, 90^\circ)$, as does β, and tan is one-to-one on that interval. So $\beta = 2\alpha - 90^\circ$, that is,

$$2\alpha = \beta + 90^\circ.$$

In words, the best direction of projection bisects the angle between the line OP and the upward vertical.

Part (ii)

At P the horizontal velocity is still $u\cos\alpha$. The time taken is $d/(u\cos\alpha)$, so the vertical velocity is

$$v_y = u\sin\alpha - \frac{gd}{u\cos\alpha}.$$

If the particle is moving at angle θ above the horizontal at P, then

$$\tan\theta = \frac{v_y}{u\cos\alpha} = \tan\alpha - \frac{gd}{u^2\cos^2\alpha} = T - \frac{gd(1+T^2)}{gdT} = T - \frac{1+T^2}{T} = -\frac{1}{T} = -\cot\alpha,$$

using $u^2 = gdT$ from part (i). So $\tan\theta = \tan(\alpha - 90^\circ)$, and since θ lies between -90° and 90°, $\theta = \alpha - 90^\circ$.

The particle is moving *downwards* as it passes through P, at an angle of $90^\circ - \alpha$ below the horizontal. Its direction at P is perpendicular to its direction of projection.

Checks

- Take $\beta = 30^\circ$. Then $\alpha = 60^\circ$ and $u^2 = \sqrt{3}\,gd$. Directly, $f(T)$ at $T = \sqrt{3}$ is $4/(\sqrt{3} - \frac{1}{\sqrt{3}}) = 2\sqrt{3}$, which matches $2T$. The final direction is 30° below the horizontal, as in the diagram.
- Letting $\beta \to 0$ gives $\alpha \to 45^\circ$ and $u^2 \to gd$. This is the familiar fact that the greatest range on level ground, u^2/g, comes from projecting at 45°. The particle then lands at 45° below the horizontal, again perpendicular to its launch direction.
- A numerical search over α for 200 random choices of d and β found the minimising angle at $\frac{1}{2}(\beta + 90^\circ)$ every time, with minimum value $gd\tan\alpha$.

Common slips

- Dropping the condition $T > k$ and keeping the root $k - \sqrt{k^2+1}$. That root is negative, which would mean projecting downwards.
- Concluding $\beta = 2\alpha - 90^\circ$ from equal tangents without saying why both angles lie in an interval where tan is one-to-one.
- Giving the angle in (ii) as $90^\circ - \alpha$ without saying that the particle is descending.

Beyond STEP

For a fixed speed u, the points you can reach lie inside a parabola, the *envelope* or "parabola of safety" of the family of trajectories. The minimum-speed trajectory to P is the one for which P lies on that envelope, and the perpendicularity in (ii) is one of that curve's geometric properties. Envelopes of families of curves, and optimisation with constraints, are developed in first-year courses. Kleppner and Kolenkow's *An*

Introduction to Mechanics is a good place to see projectile problems handled with this kind of care.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.